Foundations

Data Can Deceive · Causal Roadmaps with (and without) AI

Module C

August 5, 2026

Kathryn Morrison

Association versus Intervention

Association – *seeing*

- Patterns already in the data
- Treatment and outcome move together
- Purely descriptive: $P(Y \mid A)$

Intervention – *doing*

- What happens if *we act*
- Outcome under a hypothetical change
- Requires assumptions: $P(Y \mid do(A))$

An observed association can come from:

- a genuine causal effect,
- confounding (a common cause),
- selection, measurement, or other bias.

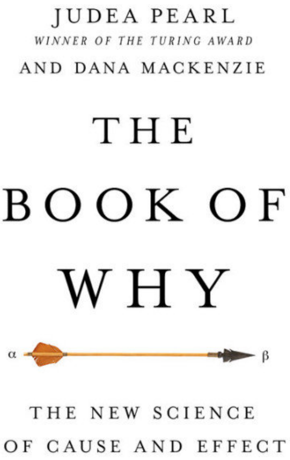
The task: move from what we *see* to what would happen if we *acted*, with the assumptions made explicit.

Pearl's Ladder of Causation

Three levels of causal reasoning [3]:

1. **Association (Seeing)**
   What does observing A tell me about Y? $P(Y \mid A)$
2. **Intervention (Doing)**
   What happens to Y if I *do* A? $P(Y \mid do(A))$
3. **Counterfactuals (Imagining)**
   What *would have happened* had A been different?

Key insight: each rung needs information the rung below cannot supply. No amount of observational data alone answers an interventional question.



The Counterfactual Contrast

A causal effect compares outcomes under different interventions:

- $Y^{(1)}$: the outcome if a person *were* treated, $A = 1$
- $Y^{(0)}$: the outcome if that same person were *not*, $A = 0$

An average causal effect exists when $E[Y^{(1)}] \neq E[Y^{(0)}]$.

Formalized in Hernán & Robins' [2] *Causal Inference: What If*.

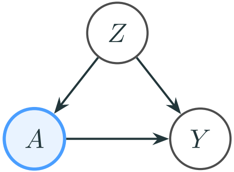
The fundamental problem of causal inference

For any one person we only ever see *one* of $Y^{(1)}, Y^{(0)}$. The other is missing, always. Causal inference is a missing-data problem we solve with design and assumptions, not with a bigger dataset.

Confounder: A Common Cause

Generic structure

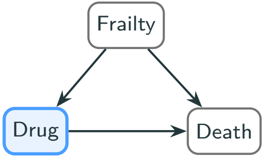
fork: $Z \to A, \quad Z \to Y$



Adjust for Z to close the backdoor.

Example: frailty

sicker patients get the aggressive drug

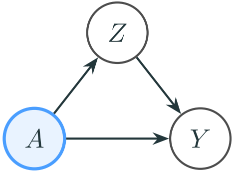


Frailty drives both treatment *and* death, so the crude contrast makes the drug look harmful.

Mediator: On the Causal Path

Generic structure

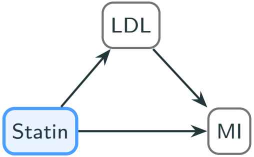
chain: $A \to Z \to Y$



Do not adjust (for a total effect).

Example: statin and LDL

the drug acts *through* lowering LDL

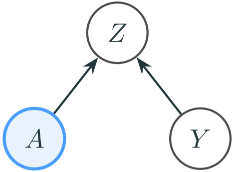


Adjust for LDL and the statin looks inert – you have removed the pathway it acts through.

Collider: A Common Effect

Generic structure

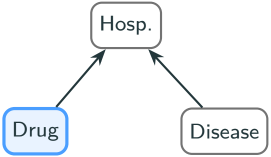
collision: $A \to Z \leftarrow Y$



Do not adjust – it opens a fake path.

Example: hospitalization

drug and disease both cause admission



Restrict to the hospitalized and drug and disease look linked.

Why “Adjust for Everything” Fails

A single reflex applied to all three does two kinds of damage:

Adjust a mediator

You block part of the effect you are trying to measure. Condition on LDL and the statin looks inert – you have thrown away the mechanism.

Adjust a collider

You *open* a path that was closed. Restrict to hospitalized patients and drug and disease look linked when in the population they are not.

The lesson: the covariate list is a set of *claims about structure*, not a bigger-is-safer knob. What to adjust for is decided by the causal question and the diagram, before the data.

Which is exactly what Segment D’s roadmap makes systematic.

Design First: The Target Trial

Specify the randomized trial you *wish* you could run [1], then emulate it with the data you have:

- eligibility, treatment strategies, assignment,
- follow-up, outcome, and the causal contrast.

Time zero: align eligibility, treatment, and follow-up at one instant

Every patient's clock starts when they meet eligibility and treatment is assigned. Getting this wrong is the classic source of **immortal-time bias** (survival smuggled into the treated group) and **prevalent-user bias** (survivors of early treatment overrepresented).

Why it matters here: a clean time zero is a *design* fix for problems no downstream model can undo.

The Pneumonia-Vaccine Paradox

Observational studies of the pneumococcal vaccine in older adults have reported that **vaccinated patients had *higher* pneumonia risk** than the unvaccinated – the vaccine looks useless, even harmful.

But look at the timing:

- In a large primary-care cohort, the vaccinated already had higher pneumonia rates *in the year before they were vaccinated*.
- An "effect" that precedes the exposure cannot be caused by it.
- The gap is a marker of *who* gets vaccinated, not of harm.

Resolution: confounding by indication

The vaccine is targeted at frailer, higher-risk patients. Baseline frailty is a **confounder** – it raises both the chance of vaccination *and* the risk of pneumonia. When the trial is actually run, the vaccine protects (CAPITA: 46% against vaccine-type pneumonia) [4].

We do not solve this with a better model first.

We solve it by designing the question.

Poll Bank

Vote in chat: A, B, or C

Poll 1 – Classify the Variable

In a study of a new anticoagulant versus an older one, **baseline kidney function** predicts both which drug the physician prescribes and the risk of a bleeding event.

Baseline kidney function is a:

A. Confounder

B. Mediator

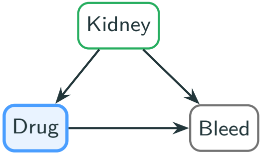
C. Collider

Poll 1 – Solution

A – Confounder.

Baseline kidney function is measured *before* treatment and causes both the drug choice and bleeding – a common cause (fork).

Action: adjust to close the backdoor.



green = adjust for it

Poll 2 – Classify the Variable

A blood-pressure drug reduces stroke risk. You have a measurement of **on-treatment blood pressure at 6 months**. You want the *total* effect of the drug on stroke.

On-treatment blood pressure is a:

A. Confounder – adjust for it

B. Mediator – do not adjust

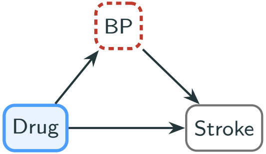
C. Collider – do not adjust

Poll 2 – Solution

B – Mediator.

On-treatment blood pressure lies on the path (drug $\to$ BP $\to$ stroke) and is measured *after* time zero.

Do not adjust for a total effect.



red box = do not condition

Poll 3 – Classify the Variable

Among **hospitalized** patients only, a drug's side effects and the underlying disease both raise the chance of admission. You analyze within the hospitalized group.

Hospitalization (the variable you restricted on) is a:

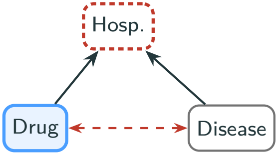
- **A.** Confounder
- **B.** Mediator
- **C.** Collider

Poll 3 – Solution

C – Collider.

Side effects and disease both cause admission, so both arrows point *into* hospitalization.

Restricting to it opens a fake link.



dashed = induced association

Poll 4 – The Tricky One

A variable Z is measured *before* treatment. It is caused by two unmeasured factors: one that also causes treatment, and one that also causes the outcome. Z itself does *not* cause A or Y.

Should you adjust for Z?

A. Yes – it is pre-treatment, so it is safe

B. No – adjusting opens a biasing path

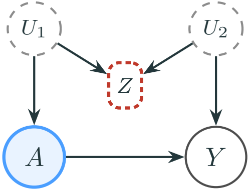
C. It never matters either way

Poll 4 – Solution

B – do not adjust (M-bias).

Z is a collider between two unmeasured causes. Conditioning on it opens the $U_1 \cdots U_2$ path and biases $A \to Y$.

"Pre-treatment" $\neq$ "safe."



dashed circle = unmeasured

Poll 5 – What Is Wrong With This Comparison?

Patients who filled a prescription for the drug *within the first year* are labelled “treated”; everyone else is “untreated.” Survival is compared from cohort entry.

The main problem is:

A. Too small a sample

B. Immortal time – the treated had to survive long enough to fill it

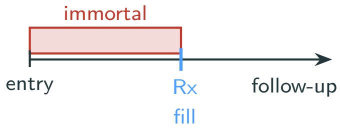
C. Measurement error in the outcome

Poll 5 – Solution

B – Immortal time.

To be labelled "treated," a patient must survive to the fill date. That guaranteed window biases toward the drug.

Fix: start time zero at treatment assignment.



survival in the shaded window is guaranteed

Poll 6 – Is This Causal?

A study finds that older adults who received the **pneumonia vaccine** have *higher* rates of pneumonia than those who did not.

The most likely explanation is:

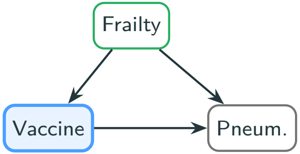
- **A.** The vaccine causes pneumonia
- **B.** Sicker, higher-risk patients were the ones vaccinated
- **C.** Random chance

Poll 6 – Solution

B – confounding by indication.

Frailer patients are targeted for the vaccine, so frailty raises both vaccination and pneumonia risk.

Tell: the gap predates vaccination.



frailty confounds the crude contrast

Poll 7 – Which Is Safe to Adjust For?

In a statin study you could adjust for LDL cholesterol measured *at baseline*, or LDL measured *at 6 months on treatment*. You want the **total** effect of the statin on heart attacks.

Which is safe to adjust for?

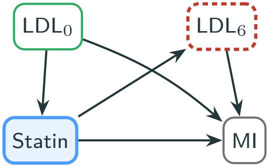
- **A.** Baseline LDL
- **B.** On-treatment LDL
- **C.** Both, to be safe

Poll 7 – Solution

A – baseline LDL only.

Baseline LDL is a pre-treatment confounder (adjust). On-treatment LDL is a mediator (do not).

Same variable, two roles – set by timing.



green=adjust red=don't

Poll 8 – The Paradox

Pooled together, patients on the new drug have *higher* mortality than those on the old drug. But within *every* disease-severity stratum, the new drug has *lower* mortality.

Which comparison reflects the drug's effect?

A. The pooled comparison

B. The within-severity comparison

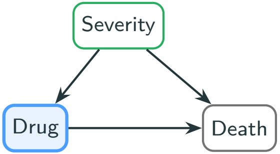
C. Neither can be trusted

Poll 8 – Solution

B – within-severity, *here*.

Severity is a pre-treatment confounder: sicker patients got the new drug. Stratifying removes the bias.

Catch: only because the stratifier is a confounder.



stratify on a confounder, not a mediator

Poll 9 – The Placebo Patients

In a randomized trial, patients who took their **placebo** as prescribed had substantially *lower* mortality than patients who did not adhere to the placebo.

The best explanation is:

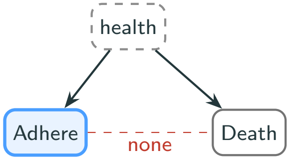
- **A.** The placebo had a real effect
- **B.** Adherence is a marker of healthier behaviour overall
- **C.** The randomization failed

Poll 9 – Solution

B – healthy-adherer bias.

Adhering even to placebo marks healthier, more health-conscious people. Adherence is a confounded proxy, not a cause.

Randomization protects the *assigned* contrast only.



unmeasured health drives both

Poll 10 – The Painkiller and the Cancer

A painkiller is prescribed for early back pain. Months later, some of those patients are diagnosed with spinal cancer – which had been causing the pain all along. In the data, the painkiller is associated with cancer.

This is:

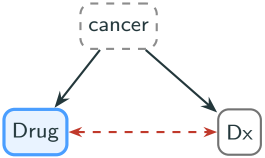
- **A.** A real carcinogenic effect
- **B.** Reverse causation: early symptoms of the outcome triggered the drug
- **C.** Confounding by age

Poll 10 – Solution

B – reverse causation.

Undiagnosed cancer caused the pain that led to the drug. The arrow runs outcome $\rightarrow$ exposure.

Fix: a lag/exclusion window before time zero.



the "effect" is driven by latent disease

Poll 11 – The Table 2 Fallacy

A model estimates a drug's effect on stroke, adjusting for age, smoking, and hypertension. The results table also reports the **smoking** coefficient and the paper reads it as "the effect of smoking on stroke."

Is that smoking coefficient a valid causal effect of smoking?

A. Yes – it was adjusted for the same covariates

B. Not generally – the adjustment set was chosen for the drug, not for smoking

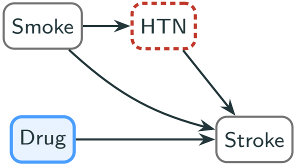
C. Yes, always

Poll 11 – Solution

B – not generally valid.

The set that identifies the *drug* effect need not identify smoking's: adjusting HTN (a mediator of smoking) blocks part of its own effect.

One model, one estimand.



adjusting HTN: right for Drug, wrong for Smoke

Poll 12 – The Feedback Loop

In an HIV study, CD4 count is *affected by* prior antiretroviral treatment, *predicts* whether treatment continues, and *predicts* mortality. You want the effect of sustained treatment on mortality.

Standard regression adjusting for CD4:

A. Works fine – CD4 is a confounder, so adjust for it

B. Is biased – CD4 both confounds later treatment *and* is affected by earlier treatment

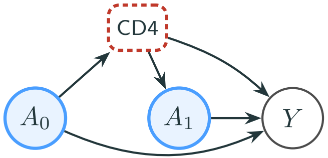
C. Should ignore CD4 entirely

Poll 12 – Solution

B – time-varying confounder.

CD4 is affected by early treatment *and* confounds later treatment. Adjust and you block/collide; omit and you confound.

Use g-methods.



CD4: mediator of A_0, confounder of A_1

Poll 13 – Spot the Instrument

You want an **instrumental variable** for whether a patient receives the new drug. Which candidate is the most plausible?

A. Baseline disease severity

B. The prescribing physician's general preference for the new drug

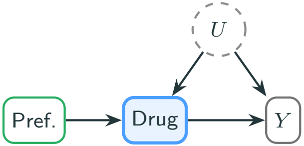
C. The patient's on-treatment lab values

Poll 13 – Solution

B – physician preference.

An instrument moves treatment but reaches Y *only through* it. Severity is a confounder; labs are a mediator.

Assumptions are strong, untestable.



instrument sidesteps unmeasured U

Poll 14 – No Comparison Group

Clinical guidelines mandate the drug for *everyone* with severe disease, so in your data there are **zero untreated** severe patients.

For the severe subgroup, you:

A. Can still estimate the effect by adjusting

B. Cannot estimate it – there is no comparison group

C. Should impute the missing untreated patients

Poll 14 – Solution

B – positivity violation.

Identification needs both arms within each stratum. With no untreated severe patients, that effect is not in the data.

Imputing the arm is extrapolation.

	Treated	Untreated
mild	• • •	• •
severe	• • •	(none)

no untreated severe patients = no contrast

Poll 15 – The Completers Analysis

In a cohort, treated patients who do poorly tend to drop out early, while untreated patients who do poorly stay in. You analyze only the patients who **completed** follow-up.

Restricting to completers:

A. Is fine – completers have the cleanest data

B. Induces selection bias – completion is a collider

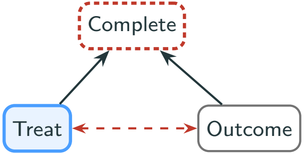
C. Only reduces power

Poll 15 – Solution

B – collider selection bias.

Completing follow-up is a common effect of treatment and prognosis. Keeping only completers conditions on that collider.

Same structure as Poll 3.



conditioning on completion opens a path

Poll 16 – Screening "Adds a Year"

Patients whose cancer is found by **screening** survive on average 3 years from diagnosis; those found after **symptoms** survive 2 years. Screening appears to add a year of life.

The most likely explanation is:

A. Screening genuinely extends life by a year

B. Lead-time bias – earlier diagnosis just starts the clock sooner

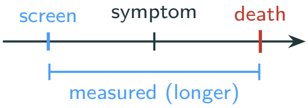
C. The screened cancers were less aggressive

Poll 16 – Solution

B – lead-time bias.

Screening moves the diagnosis date earlier. Survival *from diagnosis* stretches even if death never moves.

An artefact of where the clock starts.



earlier start, same death date

References

[1] Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8):758–764, 2016.

[2] Hernán MA, Robins JM. *Causal Inference: What If*. Chapman & Hall/CRC, 2020.

[3] Pearl J, Mackenzie D. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.

[4] Bonten MJM, Huijts SM, Bolkenbaas M, et al. Polysaccharide conjugate vaccine against pneumococcal pneumonia in adults. *New England Journal of Medicine*, 372(12):1114–1125, 2015.